

Derek Wu

Senior Backend Engineer

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Profile

Senior Backend Engineer with over 5 years of professional experience specializing in building robust Python microservices, navigating complex legacy systems, and optimizing cloud infrastructure phase-dependently. Focuses on engineering pragmatism and team-oriented synchronization over technical hype.

Immediate Legal Availability: Currently based in Hiroshima and holding a valid Working Holiday Visa through December 2026. Prepared to relocate to Tokyo within 1–2 months to join production sprints immediately with zero administrative delay or legal sponsorship risks.

Infrastructure & Cost Pragmatism: Proven track record of aligning system design with business runways; collaborated with the team to systematically migrate infrastructure from a managed cluster to streamlined dockerized VMs, safely reducing cloud costs by 53%.

Legacy Code Navigation: Highly confident in tracing and reverse-engineering dense, undocumented base-code autonomously. Successfully refactored tightly coupled legacy systems under tight timelines to unblock feature velocity for the engineering peers.

Open Source Commitment: Core creator of distributed tools and libraries garnering over 2.6k combined GitHub stars. Maintains problem-solving hand-feel through disciplined daily algorithmic practice in Rust.

Highlighted Projects

omnizart



Creator & Core Maintainer Python, TensorFlow, Signal Processing

- Built a unified, production-ready audio intelligence toolbox showcasing near **2,000 GitHub stars**; translated complex laboratory code into clean, modular library architectures consumed worldwide.

oemer



Creator & Core Maintainer Python, Computer Vision, Semantic Segmentation

- Architected and open-sourced one of the earliest end-to-end Optical Music Recognition system (**730+ GitHub stars**), successfully engineering complex digital signal processing pipelines for computer vision model inference.

Experience

03/2025 – Present



Independent Software Engineer & Technical Explorer, *Taiwan & Japan*

- **Japan Relocation:** Moved to Japan under a Working Holiday Visa; built daily and professional communication foundations, obtaining JLPT N3 qualification in 07/2025.
- **Local Workplace Adaptation:** Worked in a fast-paced environment at GALA Resort (12/2025 – 03/2026), adapting successfully to a structured Japanese team and workplace communication.
- **Rust Technical Practice:** Maintained backend development edge through hands-on pet projects: migrated a legacy Python tool to Rust (thsr-ticket-rs), explored async APIs (bfx-rs), and implemented multiple algorithms to benchmark performance across big-number crates (calc_pi, BigBench).
- **Community Engagement:** Continued volunteering for PyCon infrastructure and producing developer technical content (PyCast podcast), keeping up with ecosystem patterns.

Experience (continued)

- 12/2022 – 02/2025  **Backend & AI Engineer, *Transferhelper***
- Led infrastructure migration from Kubernetes to Docker-based virtual machines, reducing cloud infrastructure costs by 53%.
 - Architected and executed algorithmic lending strategies on Bitfinex, consistently yielding a 15% APR over a six-month period.
 - Engineered backend microservices utilizing PostgreSQL, Redis, and Kubernetes on Google Cloud Platform (GCP).
 - Established system observability pipelines through Prometheus, Grafana, and Slack alert integrations.
 - Automated CI/CD pipelines via Google Cloud Build, achieving unit test coverage exceeding 90%.
 - Researched and implemented deep learning pipelines for reconstructing 3D models from 2D human photographs.
- 01/2022 – 08/2022  **Backend Engineer, *Pinkoi***
- Led the architectural refactor and major upgrade of a legacy coupon logic system, effectively reducing code complexity and improving system performance.
 - Partnered with the cross-functional finance team to design and deploy a new automated invoice billboard feature.
 - Demonstrated strong tracing capabilities to rapidly navigate complex codebases, isolating critical production bugs and delivering precise hotfixes under tight timelines.
- 07/2021 – 11/2021  **Backend Engineer, *Meteo Piano***
- Proposed, developed, and open-sourced the earliest end-to-end Optical Music Recognition (OMR) solution, frameworked as “*oemer*” on GitHub.
 - Integrated disparate machine learning components into a highly available, distributed RESTful API server within a strict one-month deadline.
 - Deployed and managed scalable services on AWS cloud infra, utilizing EC2, S3, VPC, ECS, and Application Load Balancers.
- 06/2017 – 12/2020  **Research Assistant, *IIS Academia Sinica***
- Advanced state-of-the-art research in note-level multi-instrument music transcription, publishing two IEEE conference papers and authoring one IEEE journal paper.
 - Consolidated isolated laboratory research models into a unified, production-ready Python package, *omnizart*, which garnered near 2,000 GitHub stars.
- 07/2019 – 06/2020  **Backend Engineer Intern, *TrendMicro***
- Resolved a long-standing, complex cross-team bottleneck within two months by securely bridging internal ADFS authentication with AWS IAM services.
 - Optimized deployment pipelines on AWS, integrating a robust Blue/Green deployment strategy for zero-downtime releases.
 - Streamlined legacy deployment scripts for readability, implementation of unit tests, and reduced overall CI/CD runtime by up to 50%.
 - Ported high-performance machine learning execution blocks from Python to Java to ensure seamless backend infrastructure integration.

Education

- 2018 – 2020  **M.Sc. Computer Science**, National Taiwan Normal University.
Thesis title: *Multi-Instrument Automatic Music Transcription With Self-Attention-Based Instance Segmentation*.
Publications: Published 3 IEEE papers (including 1 IEEE Journal paper) on deep-learning-based music transcription models.
- 2014 – 2018  **B.Sc. Computer Science**, National Taiwan Normal University.

Skills

- Languages  Native Mandarin Chinese speaker; strong proficiency in reading and writing English.
- Coding  Python, FastAPI, Rust, asynchronous programming.
- Databases  PostgreSQL, MySQL.
- Infra  Google Cloud Platform (GCP), Amazon Web Services (AWS), Docker, Kubernetes, GitHub, Redis, Nginx, Linux.
- Misc.  Academic research, machine learning, TensorFlow, PyTorch, scikit-learn, NumPy, OpenCV.